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In [1]: import pandas as pd
import numpy as np
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In [12]: #Step 2: Load the Dataset Properly
# Load dataset
df = pd.read_csv('dataset_Facebook.csv')

# Display first 5 rows
df.head()
```

```
Out[12]:
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	pagetotallikes	type	category	postmonth	postweekday	posthour	paid	lifetimeposttotalreach
0	139441	2	2	12	4	3	0	2752
1	139441	3	2	12	3	10	0	10460
2	139441	2	3	12	3	3	0	2413
3	139441	2	2	12	2	10	1	50128
4	139441	2	2	12	2	3	0	7244

```
In [25]: #Step 3: Create Column Subset
subset_columns = df[[
    'pagetotallikes',
    'type',
    'category',
    'postmonth',
    'postweekday',
    'lifetimeposttotalreach'
]]

subset_columns.head()
```

```
Out[25]:
```

	pagetotallikes	type	category	postmonth	postweekday	lifetimeposttotalreach
0	139441	2	2	12	4	2752
1	139441	3	2	12	3	10460
2	139441	2	3	12	3	2413
3	139441	2	2	12	2	50128
4	139441	2	2	12	2	7244

```
In [26]: #Step 4: Create Row Subset
high_reach = df[df['lifetimeposttotalreach'] > 50000]

high_reach.head()
```

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Out[26]:
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	pagetotallikes	type	category	postmonth	postweekday	posthour	paid	lifet
3	139441	2	2	12	2	10	1	
17	138414	2	1	12	2	12	1	
61	138185	2	1	11	3	2	1	
67	138185	2	1	11	7	3	1	
71	137893	4	1	11	5	3	1	

```
In [27]: #Step 5: Create Type-Based Subset
photo_posts = df[df['type'] == 'Photo']

photo_posts.head()
```

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Out[27]:
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	pagetotallikes	type	category	postmonth	postweekday	posthour	paid	lifetim
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In [28]: # B Merge Data
#Step 6: Create Two DataFrames
df1 = df[['type', 'category', 'postmonth']]

df2 = df[['like', 'share', 'comment']]
```

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In [29]: #Step 7: Merge DataFrames
merged_df = pd.concat([df1, df2], axis=1)

merged_df.head()
```

```
Out[29]:
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	type	category	postmonth	like	share	comment
0	2	2	12	79	17	4
1	3	2	12	130	29	5
2	2	3	12	66	14	0
3	2	2	12	1572	147	58
4	2	2	12	325	49	19

```
In [30]: merged_df.to_csv('Merged_Facebook_Data.csv', index=False)

print("Dataset saved successfully!")
```

Dataset saved successfully!

```
In [ ]:
```